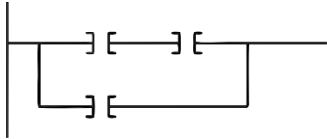


BORE LAB: INPUT CONDITIONS LIST WORKSHEET

Instructions with a single **asterisk *** are in series within the holding circuit.

STUDIO 5000



If there are no asterisks when a holding circuit is required, then there is a single instruction in parallel with the holding circuit.

COMPLETE THE INSTRUCTIONS AND TAGS

RUNG 0: SOL_CLAMP1, SOL_CLAMP2

-] [-	SAFETY_GATE	Safety gate closed.
-] [-	PART_PE	Part detected photo-electric switch.
-] [-	ENGAGE_CLAMP_SW	Switch to engage clamps.

RUNG 1: CYCLE_START_CR

-] [-	START_SIGNAL	Signal to the PLC when the pushbuttons are pressed.
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RUNG 2: CYCLE_COMPLETE_CR (Requires holding circuit)

_____ *	_____	Up control relay.
_____ *	_____	Up Limit Switch.
_____	_____	Holding Circuit
_____	_____	A control relay (internal) to break the holding circuit when the operator releases the pushbuttons. <i>This control relay is not set until the spindle carriage returns to the Up position.</i>

RUNG 3: DOWN_CR (Requires holding circuit)

_____ *

A control relay set when pushbuttons are pressed to start the cycle.

_____ *

A control relay to prevent the branch from being true after the cycle has started.

Notice the CYCLE START CONTROL RELAY is set for the entire movement of the spindle carriage, both up and down. This interlock instruction is needed to trigger the DOWN CONTROL RELAY, but once the machine is in progress the interlock prevents the machine from cycling again after returning to the top.

Holding Circuit

An input to break the holding circuit, when it senses the spindle carriage is down.

A control relay which is set to prevent the spindle carriage from moving down when the entire cycle is completed.

RUNG 4: SPINDLE_ON, SOL_DOWN

Process input for gate that must be set for operation.

Process input for clamp 1 that must be set for operation.

Process input for clamp 2 that must be set for operation.

A control relay set when pushbuttons are pressed.

Control relay which commands the down action.

Interlock of the control relay commanding the up action.

RUNG 5: UP_CR (Requires holding circuit)

_____	_____	Down limit switch.
_____	_____	Holding Circuit
_____	_____	An input to break the holding circuit, when it senses the spindle carriage is up.

RUNG 6: UP SOLENOID

_____	_____	Process input for gate that must be set for operation.
_____	_____	Process input for clamp 1 that must be set for operation.
_____	_____	Process input for clamp 2 that must be set for operation.
_____	_____	A control relay set when pushbuttons are pressed.
_____	_____	Control relay which commands the up action.
_____	_____	Interlock of the control relay commanding the down action.

RUNG 7: CYCLE_IN_PROGRESS_CR (Requires holding circuit)

_____*	_____	Down control relay.
_____*	_____	Input sensing the spindle carriage has moved from up position.
_____	_____	Holding Circuit
_____	_____	A control relay to break the holding circuit, when the entire operation is completed.